

A Lab Report on Requirement Analysis for Web-Based Alumni Networking System



Submitted By:

Prachit Bhujel [PUR079BCT050]

Pratyoos Panta [PUR079BCT054]

Sagar Mainali [PUR079BCT069]

Tribhuvan University

Institute of Engineering

Department of Electronics and Computer Engineering

Purwanchal Campus

Dharan-8, Sunsari

Objective

The primary aim of this lab session is to prepare a mini project document including the requirement analysis, problem statement, relevant software development model, and functional and non-functional requirements for a Web-Based Alumni Networking System.

Functional Overview

- Document the requirement analysis to define the system's purpose and needs.
- Outline the problem statement to identify existing gaps.
- Specify objectives to guide the project direction.
- Define scope and limitations to set project boundaries.
- Select and justify a suitable software development model.
- List functional and non-functional requirements for system development.

This report provides a comprehensive foundation for the Web-Based Alumni Networking System, ensuring all requirements are clearly documented to support subsequent design and development phases.

Requirement Analysis

PHASE 1: INTRODUCTION

1.1 Introduction

In the modern academic environment, maintaining strong connections with alumni has become essential for institutional development and student career support. We are planning to develop a web-based Alumni Networking System that will enable alumni to register, interact, and collaborate with their peers and institution.

The platform will serve as a central hub for alumni to stay engaged, contribute back, and build professional networks. It will also help current students benefit from mentorship, industry insights, and job referrals from graduates.

1.2 Problem Statement

Currently, there is no centralized system in our institution that enables alumni to connect with each other or with current students. Important alumni events and opportunities often go unnoticed due to poor communication. This leads to a disconnect between the institution and its graduates. A lack of organized platforms for networking and knowledge sharing limits the potential benefits that alumni engagement can offer. Therefore, we aim to create a structured system that brings together alumni, students, and faculty in one collaborative environment.

1.3 Objectives

Our project aims to fulfill the following objectives:

- Develop a centralized web-based platform for alumni engagement.
- Allow users to register, create profiles, and connect with peers.
- Enable secure communication between alumni and institution.
- Provide a dashboard for event listings, job posts, and announcements.
- Create an admin interface for content and user management.
- Promote collaboration between alumni and current students.

1.4 Scope and Limitations

Scope:

- The system will support alumni registration and authentication.
- Users will be able to post and view events, jobs, and announcements.
- Messaging and connection features will allow peer communication.
- Admins will be able to manage users and content.
- The platform will support mobile responsiveness.

Limitations:

- The initial version will be limited to alumni of our campus only.
- Delivery systems or physical meetups will not be handled.
- External social media integrations may not be included at first.
- Engagement will depend on user participation.

PHASE 2: SYSTEM ANALYSIS AND DESIGN

2.1 Development Methodology

We plan to use the Agile Software Development Methodology for this project. Agile is suitable for web applications that require continuous improvement, testing, and feedback. It allows us to work in iterative cycles, incorporating changes based on user suggestions and real-world usage.

Planned Phases:

- Requirement Gathering: Collecting needs and defining key features.
- Planning: Dividing the system into modules and setting timelines.
- Design: Creating wireframes, UI prototypes, and database schemas.
- Implementation: Building the project using MERN stack.
- Testing: Performing unit, integration, and user testing to identify bugs.
- Deployment: Hosting the system on a suitable cloud provider.

- Feedback: Gathering responses from users and improving the platform.

2.2 Requirement Analysis

Functional Requirements (Planned):

- Alumni registration and login functionality.
- Profile creation and editing.
- Messaging between alumni users.
- Event creation and joining options.
- Announcement board visible to all users.
- Admin control panel for user and content management.

Non-Functional Requirements (Planned):

- Performance: The system should respond quickly under standard load.
- Scalability: Must be able to support hundreds of users in future phases.
- Availability: Should maintain at least 99.9% uptime.
- Security: Data will be protected using encryption and authentication.
- Responsiveness: UI will adapt to both desktop and mobile screens.
- Maintainability: Code will be modular and well-documented.

A Lab Report on System Requirements Specification (SRS) for Web-Based Alumni Networking System



Submitted By:

Prachit Bhujel [PUR079BCT050]

Pratyoos Panta [PUR079BCT054]

Sagar Mainali [PUR079BCT069]

Tribhuvan University

Institute of Engineering

Department of Electronics and Computer Engineering

Purwanchal Campus

Dharan-8, Sunsari

Objective

The primary aim of this lab session is to prepare a System Requirements Specification (SRS) document for a Web-Based Alumni Networking System, detailing the technical, operational, and economic feasibility to ensure a robust foundation for system design and development.

Functional Overview

- **SRS Document:** Provides a detailed specification of system requirements, including feasibility analysis.
- **Feasibility Analysis:** Assesses the technical, operational, and economic viability of the project.
- **Foundation for Design:** Serves as a basis for subsequent development phases, including data flow diagrams and implementation.

This report establishes the SRS, ensuring all feasibility aspects are addressed to support the successful development of the Web-Based Alumni Networking System.

System Requirements Specification (SRS)

PHASE 2: SYSTEM ANALYSIS AND DESIGN

2.3 Feasibility Analysis

Technical Feasibility: The project will use the MERN stack (MongoDB, Express.js, React.js, and Node.js) which is widely supported and familiar to the team. Tools like Postman, GitHub, and VS Code will be used during development.

Operational Feasibility: The platform will meet institutional goals by improving alumni engagement and helping students through mentorship and connections.

Economic Feasibility: Free-tier services like Vercel and MongoDB Atlas will minimize development costs. No licensing or infrastructure costs are expected initially.

A Lab Report on Data Flow Diagrams (DFD) for Web-Based Alumni Networking System



Submitted By:

Prachit Bhujel [PUR079BCT050]

Pratyoos Panta [PUR079BCT054]

Sagar Mainali [PUR079BCT069]

Tribhuvan University

Institute of Engineering

Department of Electronics and Computer Engineering

Purwanchal Campus

Dharan-8, Sunsari

Objective

The primary aim of this lab session is to design and analyze Data Flow Diagrams (DFDs) to represent the data flow within a Web-Based Alumni Networking System. These diagrams help illustrate how data moves through the system, including processes, data stores, and interactions between users and the system, enhancing understanding and aiding in efficient system development.

Functional Overview

- **Level 0 DFD:** Provides an overview of the system, showing major processes and data flows between external entities and the system.
- **Level 1 DFD:** Breaks down the Level 0 diagram into detailed processes, depicting how data is processed and stored within the system.
- **Level 2 DFD:** Further decomposes specific processes from the Level 1 DFD to show more detailed data flows and interactions.

This report utilizes DFDs to map out the data flow, identifying key processes and data storage points. These diagrams serve as a blueprint for developers and stakeholders, ensuring a clear understanding of data movement and supporting iterative development approaches.

Diagrams

Level 0 DFD

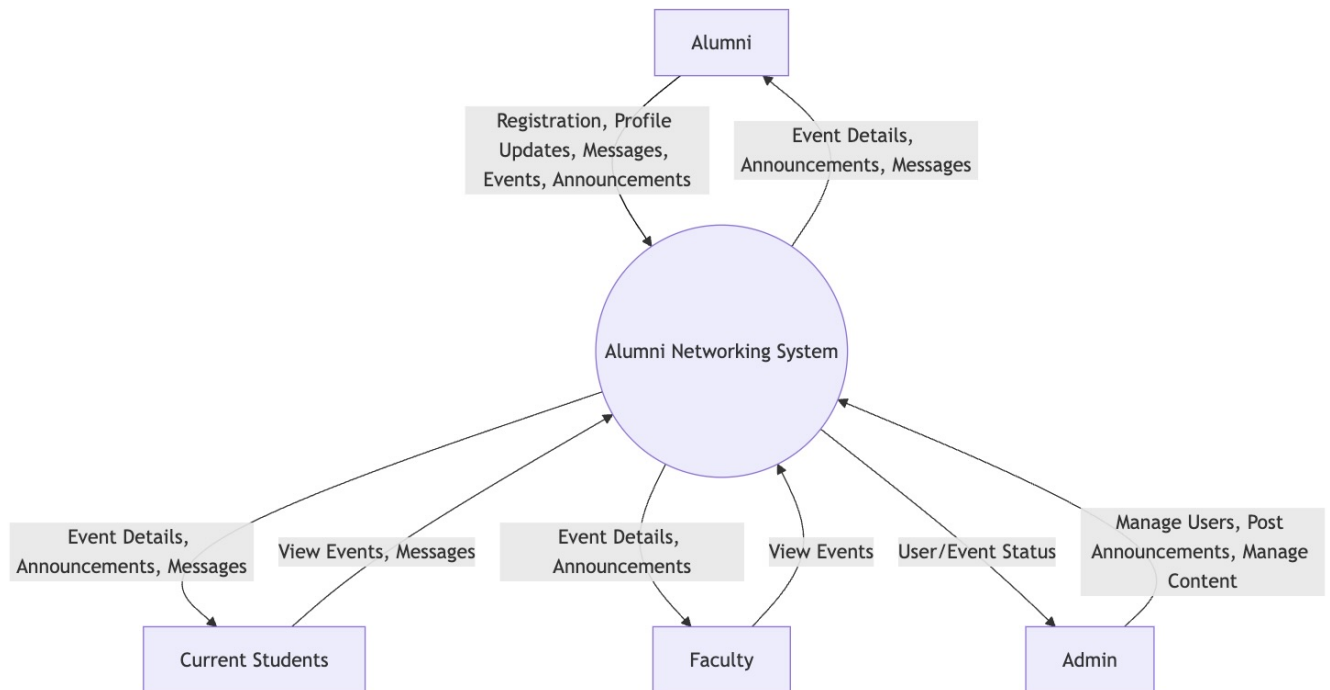


Figure 1: This Level 0 DFD provides an overview of the Alumni Networking System, showing interactions with Alumni, Admin, and Current Students. It includes major processes like managing profiles, users, messages, and events/announcements, with data flows to Alumni Database, Message Database, and Event Database.

Level 1 DFD

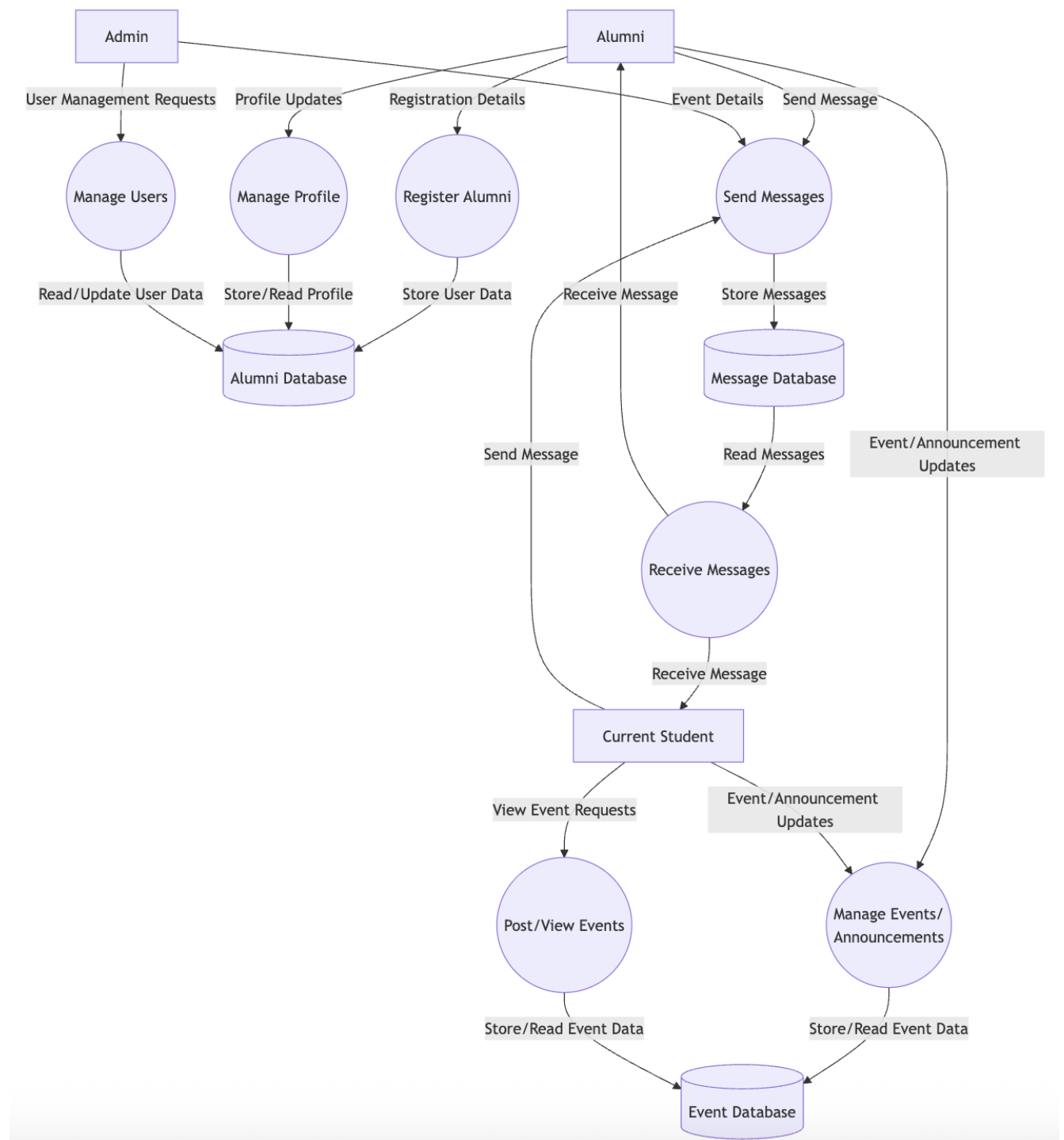


Figure 2: This Level 1 DFD expands the Level 0 diagram, detailing processes such as Register Alumni, Manage Profile, Manage Users, Send/Receive Messages, Manage Events/Announcements, and Post/View Events. It shows data flows to and from the Alumni Database, Message Database, and Event Database.

Level 2 DFD - Register Alumni Process

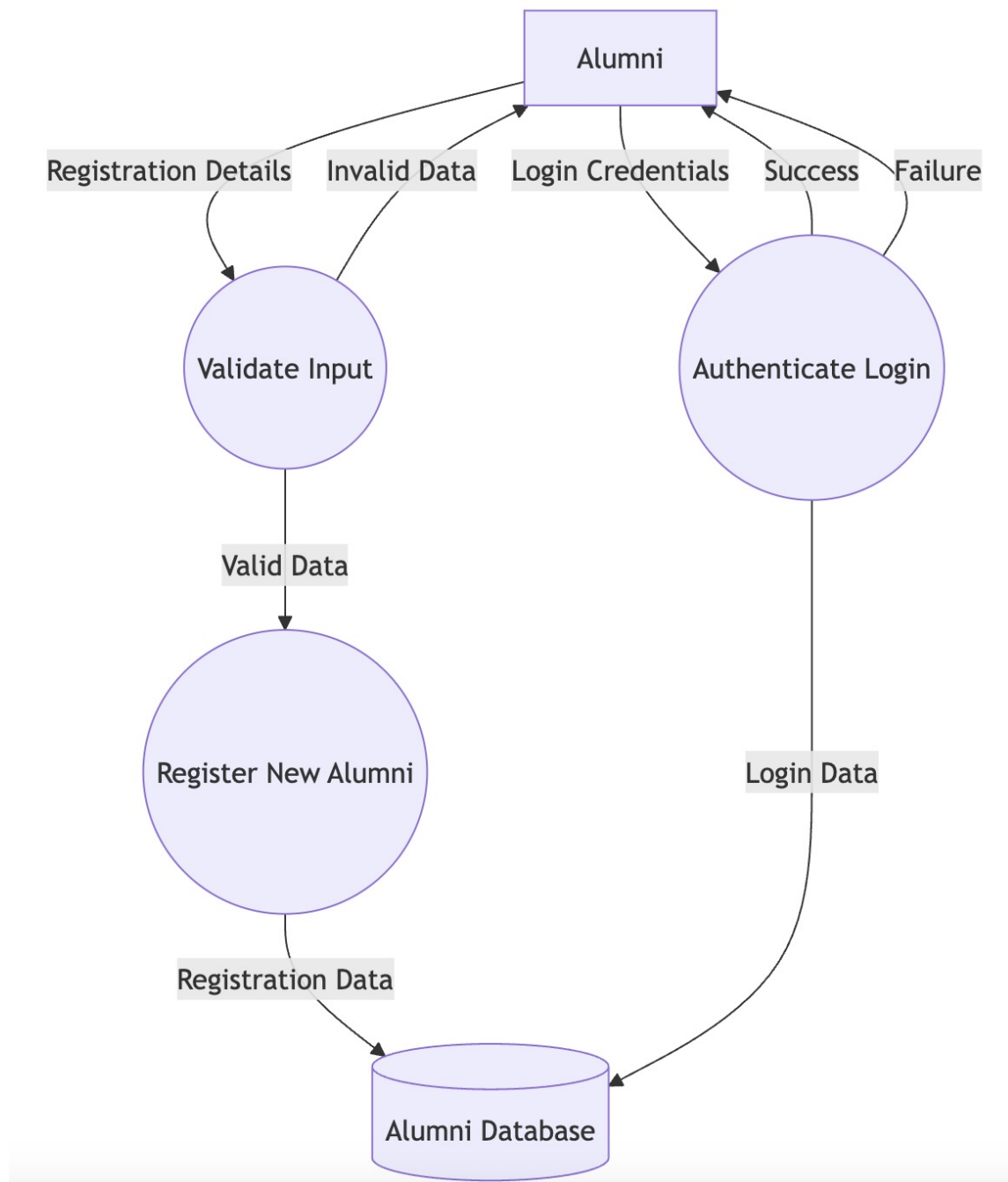


Figure 3: This Level 2 DFD focuses on the Register Alumni process, illustrating the steps of validating input and storing registration data in the Alumni Database, with data flows for registration details, valid data, and potential failures.

Level 2 DFD - Send Message Process

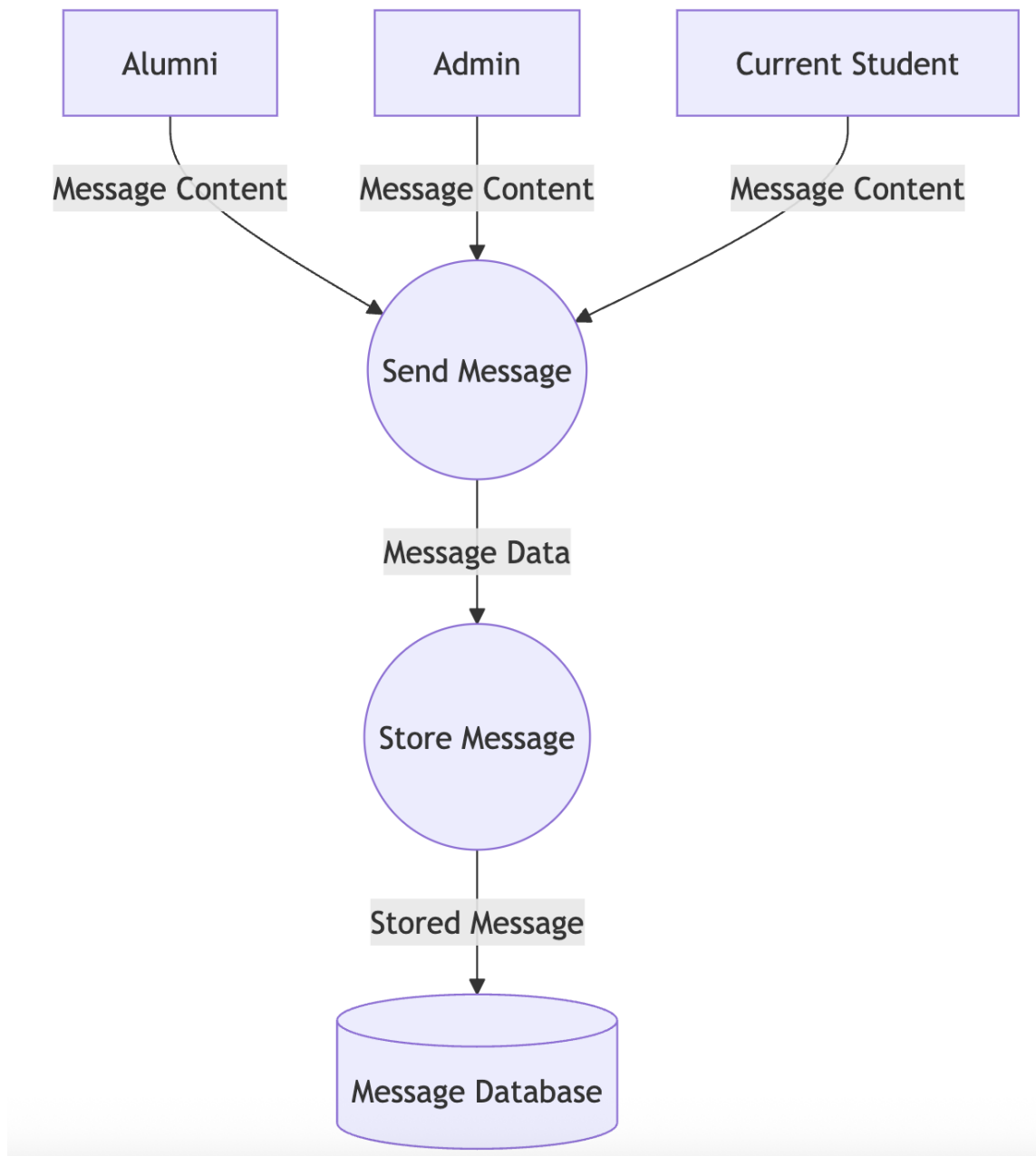


Figure 4: This Level 2 DFD details the Send Message process, showing how message content from Alumni, Admin, and Current Students is sent, received, and stored in the Message Database.

Level 2 DFD - Update Profile Process

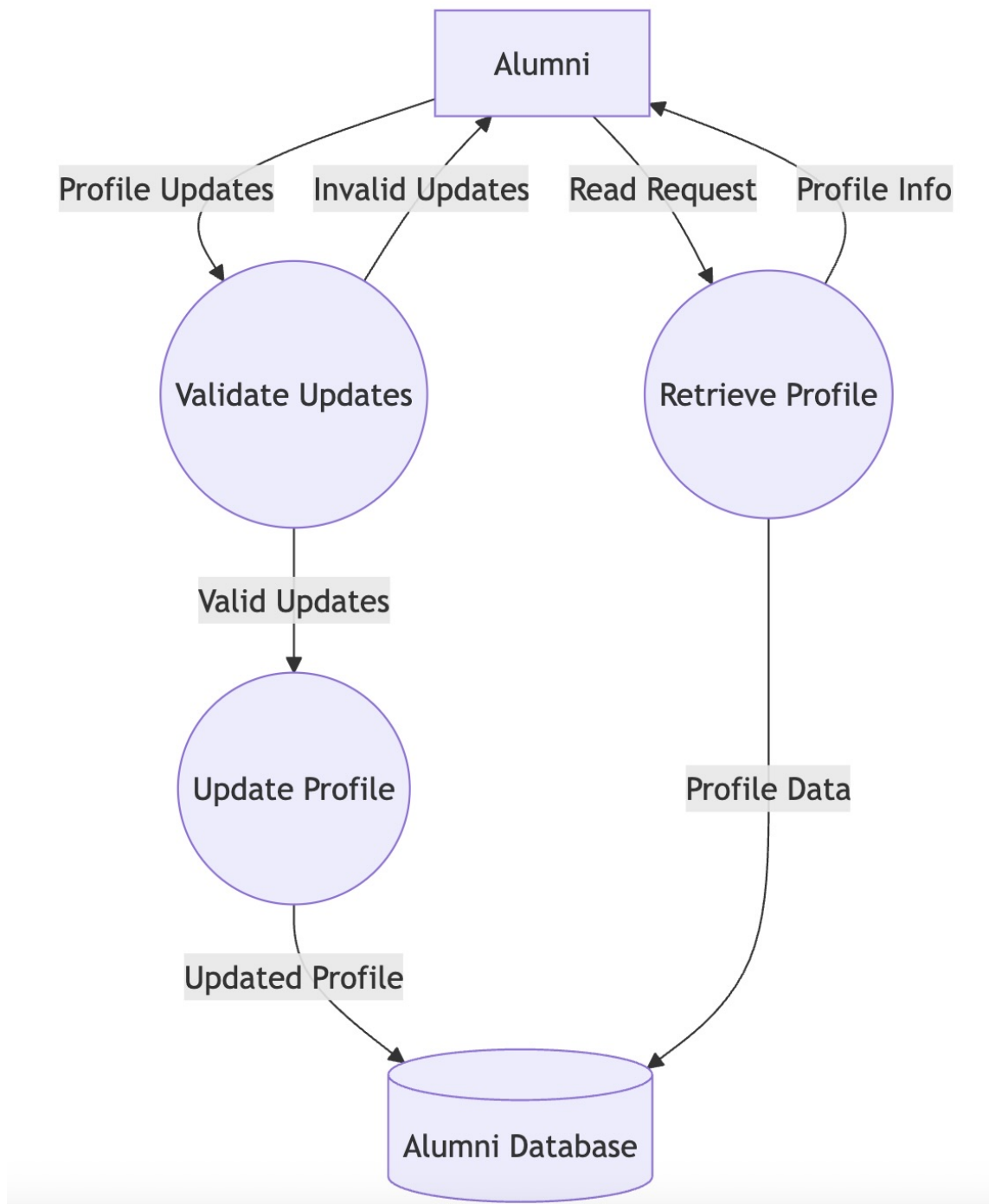


Figure 5: This Level 2 DFD illustrates the Update Profile process, depicting validation of profile updates and storage of updated profile data in the Alumni Database.

Level 2 DFD - Manage Users Process

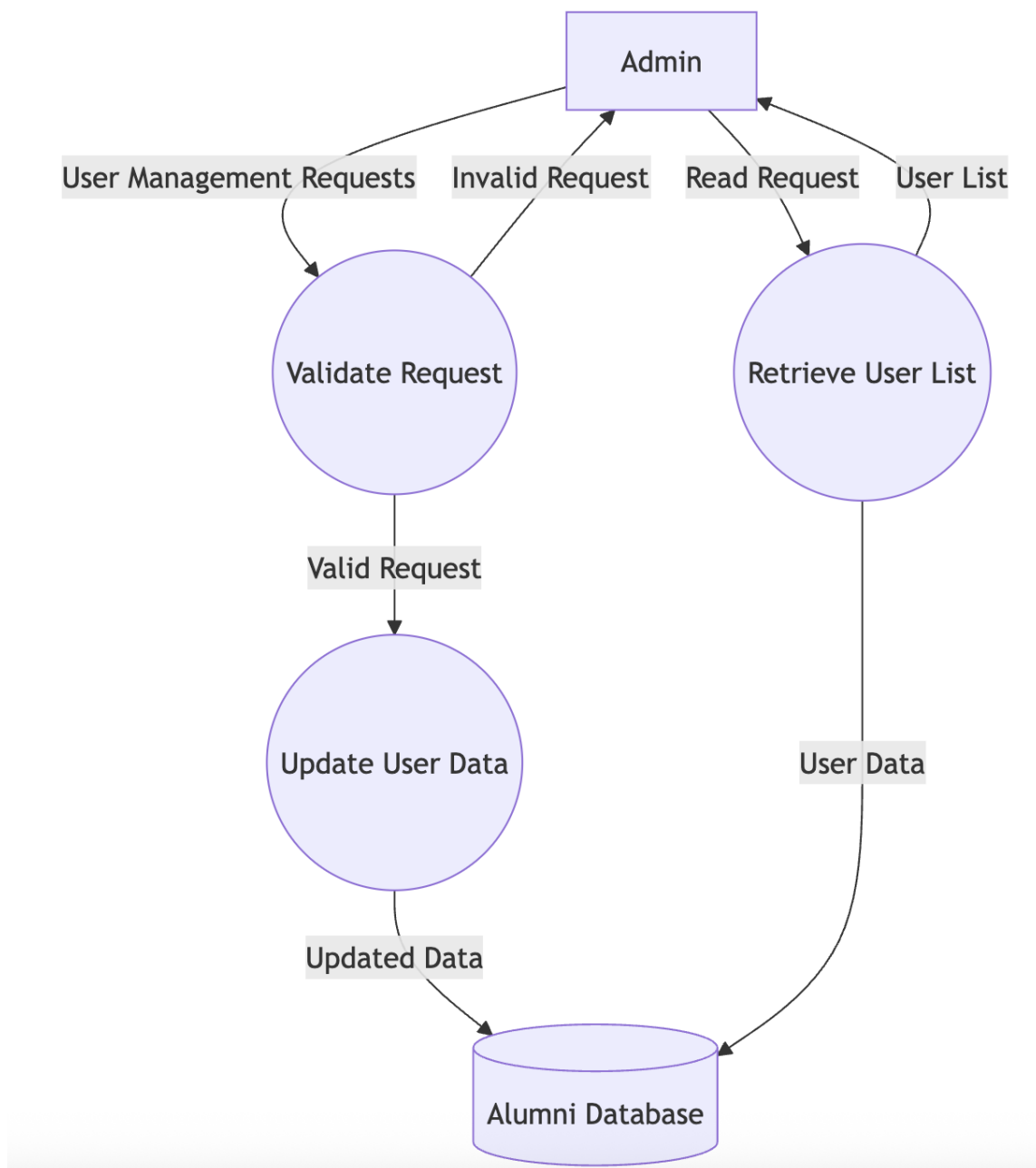


Figure 6: This Level 2 DFD details the Manage Users process, showing validation of requests and updating user data in the Alumni Database.

Level 2 DFD - Post/View Events Process

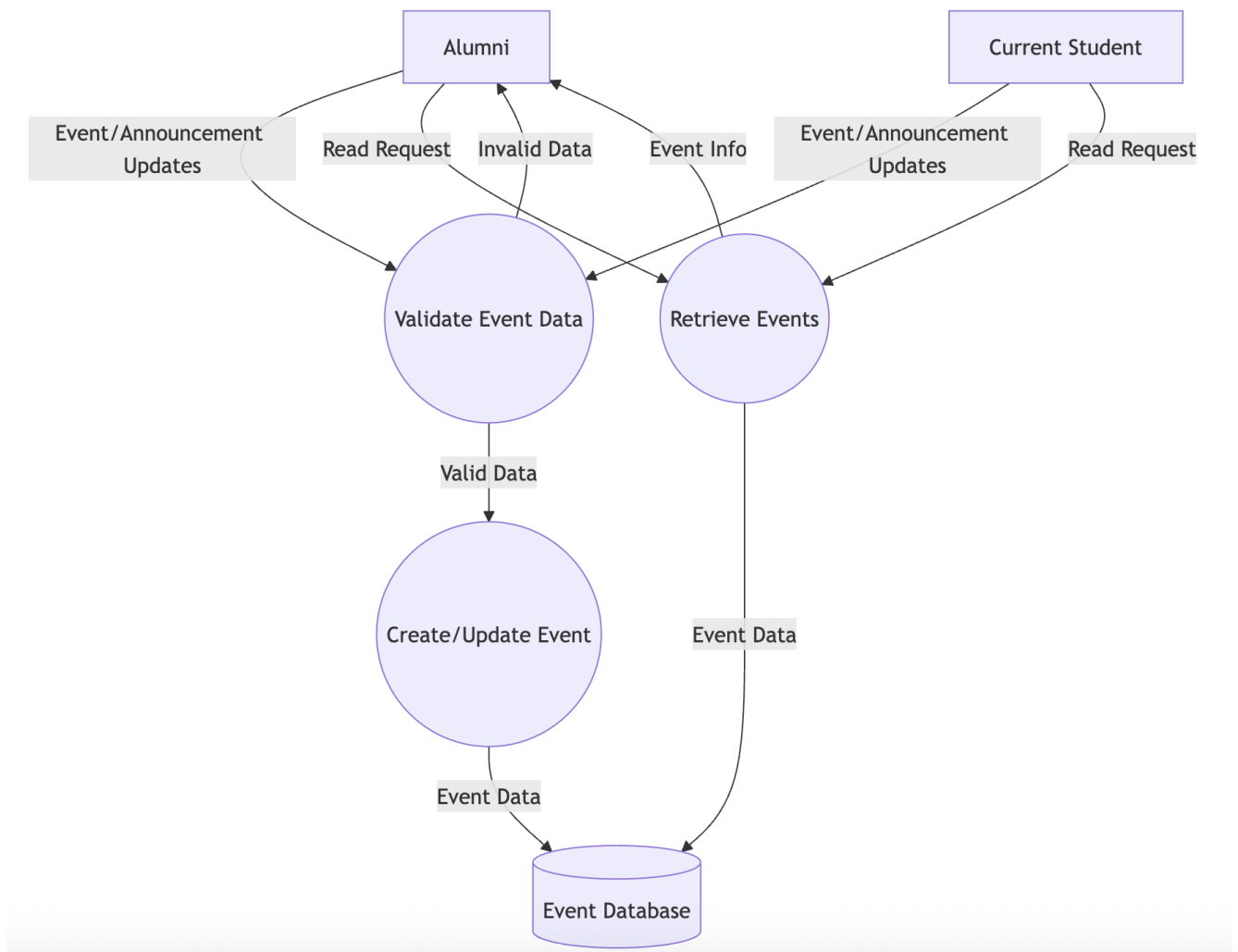


Figure 7: This Level 2 DFD outlines the Post/View Events process, illustrating event/announcement updates and data storage/retrieval from the Event Database.

A Lab Report on Use Case, Activity, Sequence, and Class Diagrams for Web-Based Alumni Networking System



Submitted By:

Prachit Bhujel [PUR079BCT050]

Pratyooos Panta [PUR079BCT054]

Sagar Mainali [PUR079BCT069]

Tribhuvan University

Institute of Engineering

Department of Electronics and Computer Engineering

Purwanchal Campus

Dharan-8, Sunsari

Objective

The primary aim of this lab session is to design and analyze Use Case, Class, Activity, and Sequence diagrams for the Web-Based Alumni Networking System to model system behavior and structure.

Functional Overview

- **Use Case Diagram:** Illustrates interactions between actors and the system.
- **Class Diagram:** Defines the static structure and relationships of system entities.
- **Activity Diagram:** Depicts the dynamic workflow of subprocesses.
- **Sequence Diagram:** Shows the time-ordered interactions between actors and the system.
- **Purpose:** Enhances understanding of system design and supports development.

This report utilizes diagrams to provide a comprehensive view of the Alumni Networking System's functionality and structure.

Diagrams

1. Use Case - System Interactions

The Use Case Diagram provides an overview of the interactions between actors (Alumni, Admin, Current Students) and the Web-Based Alumni Networking System. It includes major use cases such as Register Alumni, Manage Profile, Send/Receive Messages, Manage Users, and Post/View Events/Announcements. Each use case represents a specific functionality, with arrows indicating the direction of interaction. This diagram helps stakeholders understand the system's intended behavior and actor responsibilities.

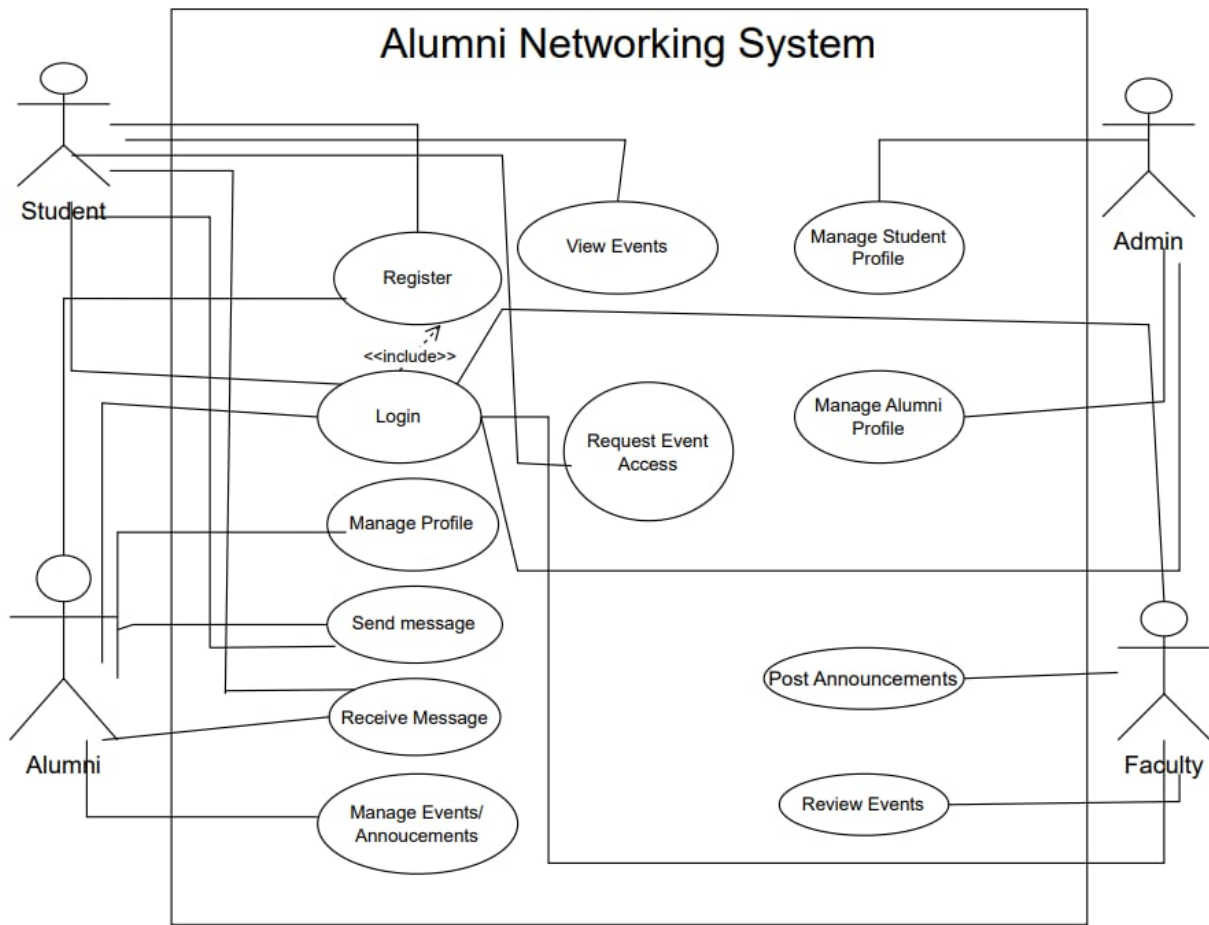


Figure 1: Use Case Diagram for Alumni Networking System

2. Class - System Structure

The Class Diagram outlines the static structure of the Alumni Networking System, defining classes such as Alumni, Admin, Event, Message, and Databases (Alumni Database, Message Database, Event Database). It includes attributes (e.g., Alumni ID, Event Title) and methods (e.g., updateProfile(), sendMessage()), along with relationships like associations (e.g., Alumni manages Event) and dependencies. This diagram serves as a blueprint for the system's object-oriented design, ensuring clarity in data and behavior.

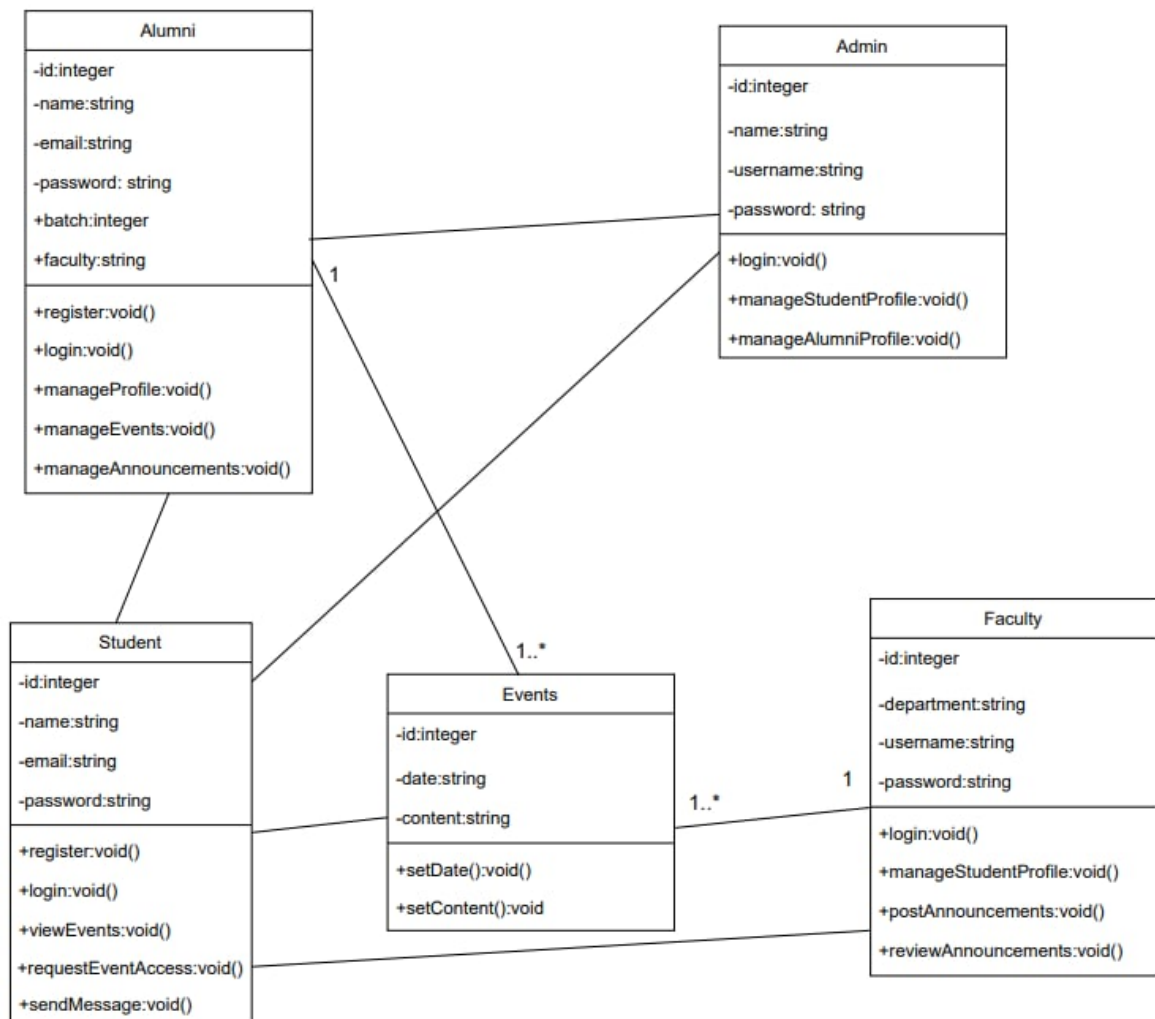


Figure 2: Class Diagram for Alumni Networking System

3. Activity - Subprocess Workflows

The following Activity Diagrams detail the dynamic workflows for each subprocess identified in the Level 2 DFDs.

3.1 Activity - Register Alumni Process

This diagram illustrates the workflow for registering an alumni, starting with receiving a registration request, validating input, storing data in the Alumni Database if valid, notifying the Admin, and ending. Invalid inputs trigger an error display, looping back for correction.

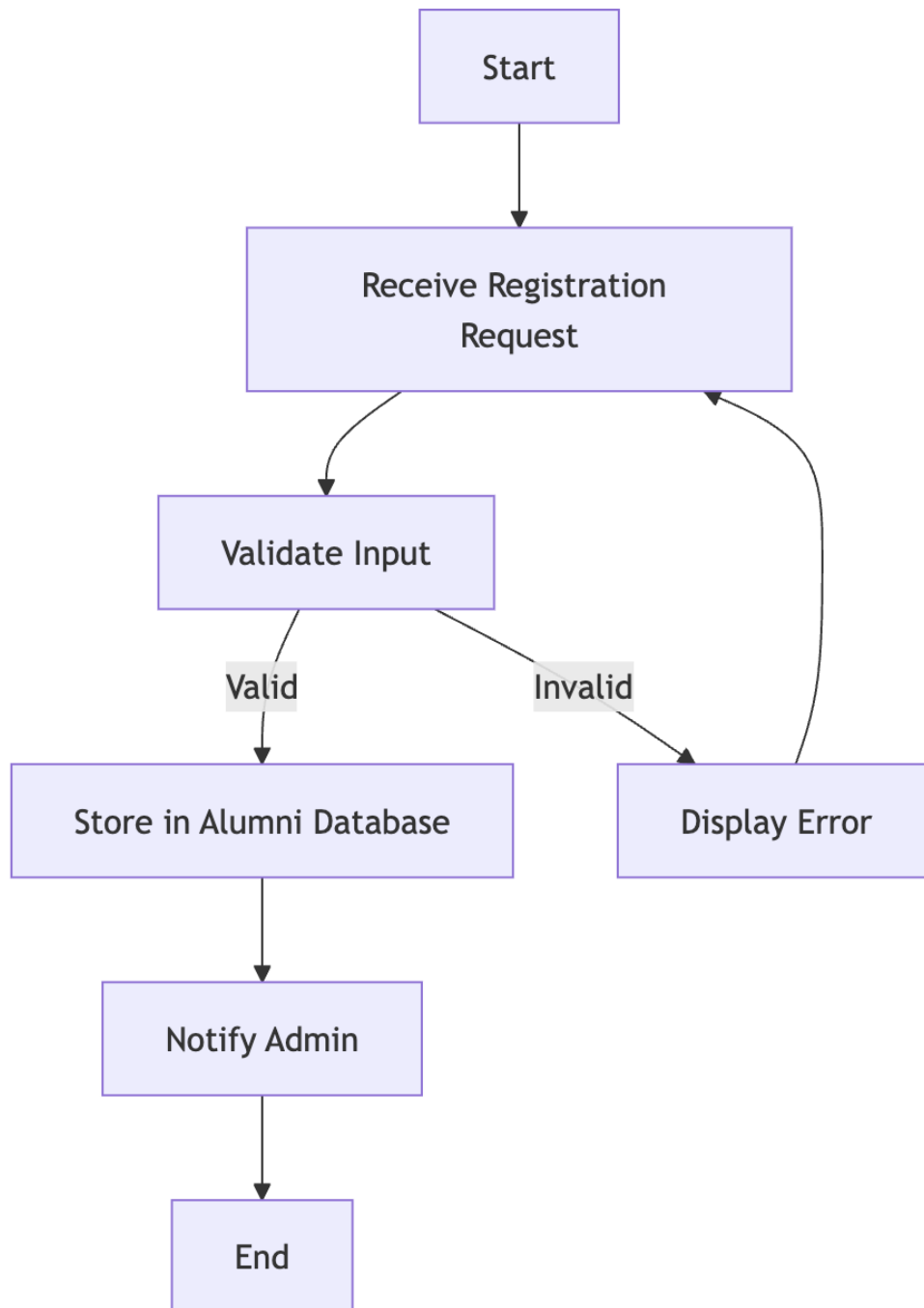


Figure 3: Activity Diagram for Register Alumni Process

3.2 Activity - Send Message Process

This diagram maps the process of sending messages, where users (Alumni, Admin, Current Students) compose a message, validate it, store it in the Message Database, deliver it to the recipient, and end. Invalid messages loop back for revision.

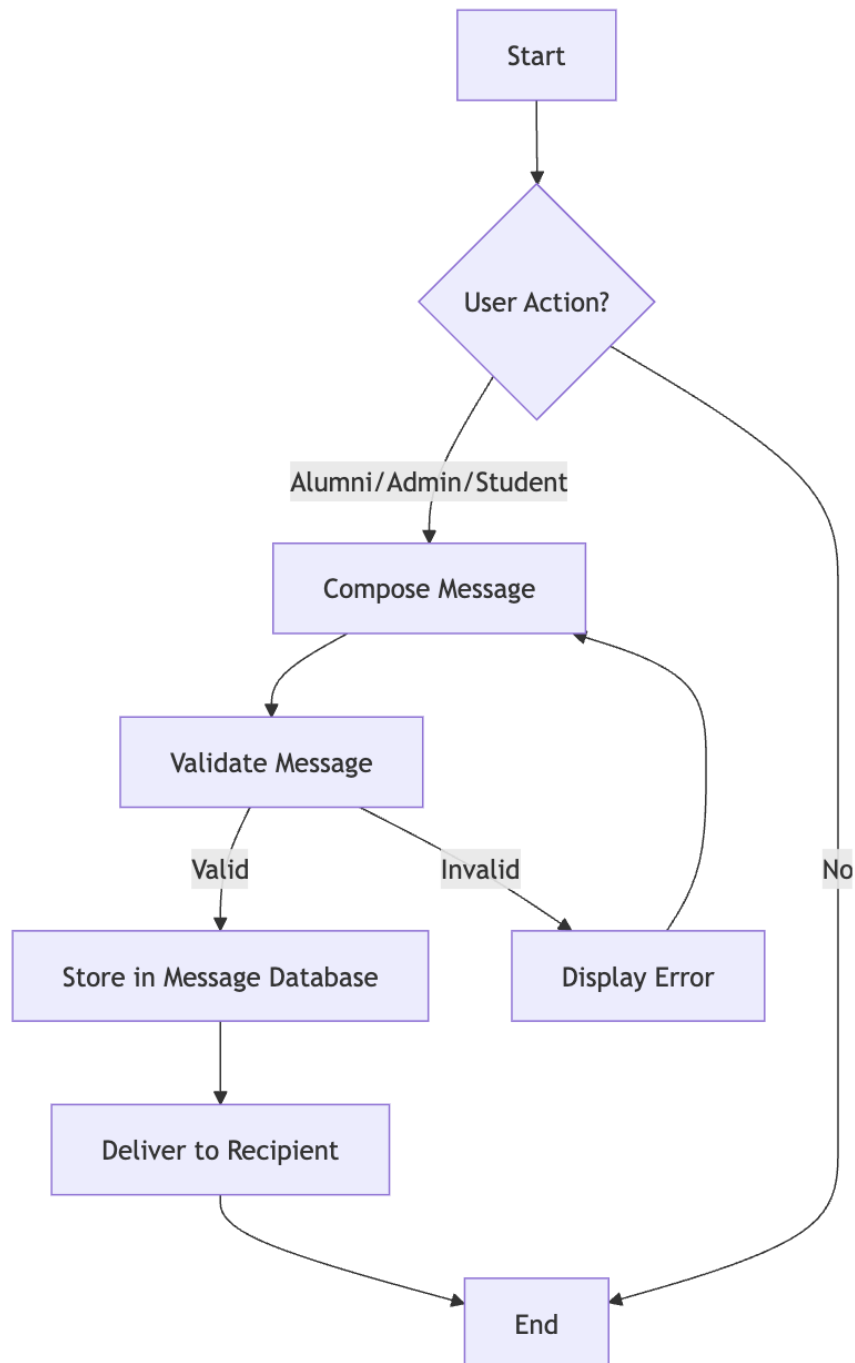


Figure 4: Activity Diagram for Send Message Process

3.3 Activity - Update Profile Process

This diagram depicts the update profile workflow, receiving an update, validating it, updating the Alumni Database if valid, notifying the user, and ending. Invalid updates return an error for re-entry.

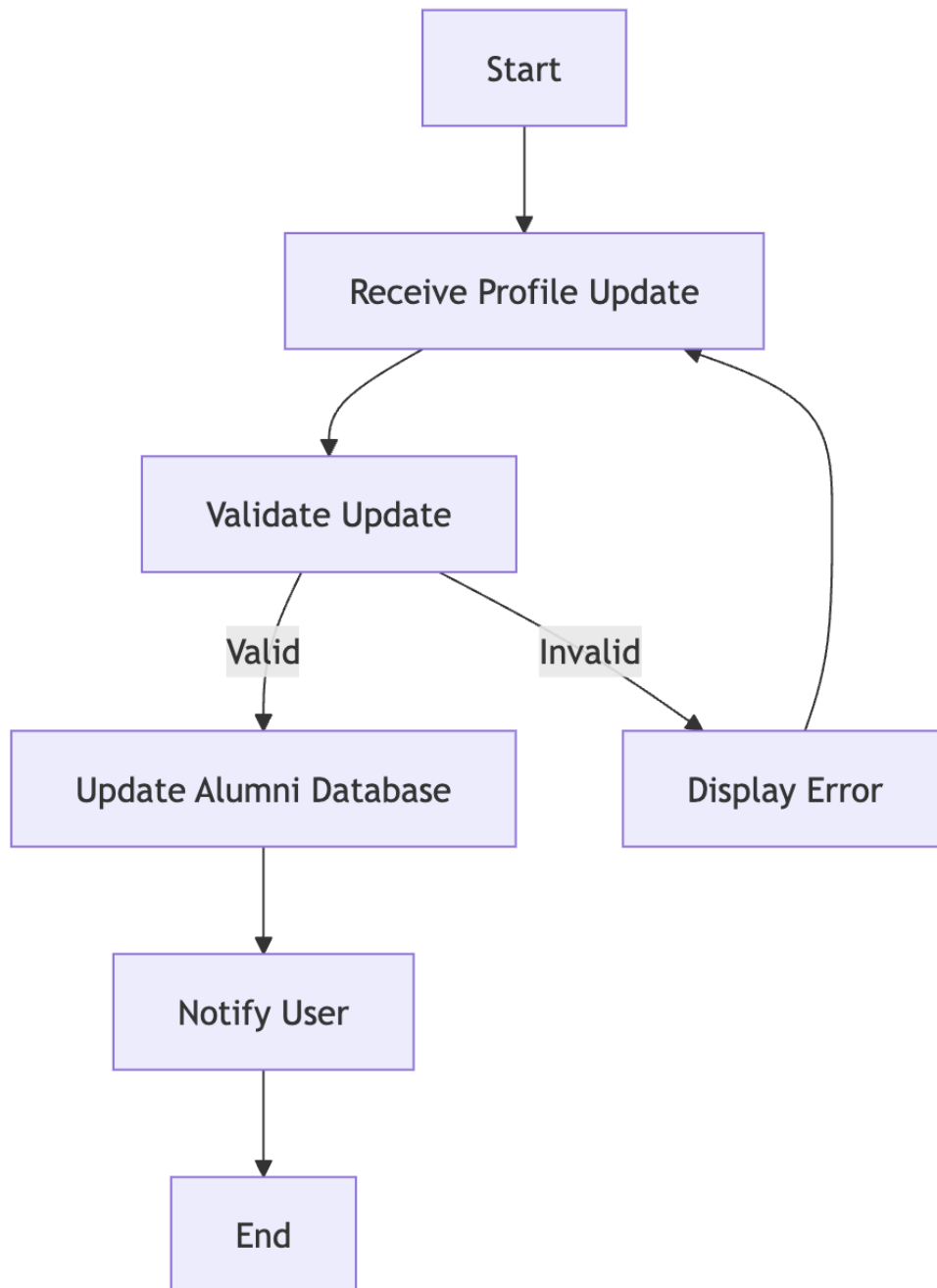


Figure 5: Activity Diagram for Update Profile Process

3.4 Activity - Manage Users Process

This diagram outlines the manage users workflow, initiated by the Admin, receiving a request, validating it, updating the Alumni Database if valid, notifying the Admin, and ending. Invalid requests loop back.

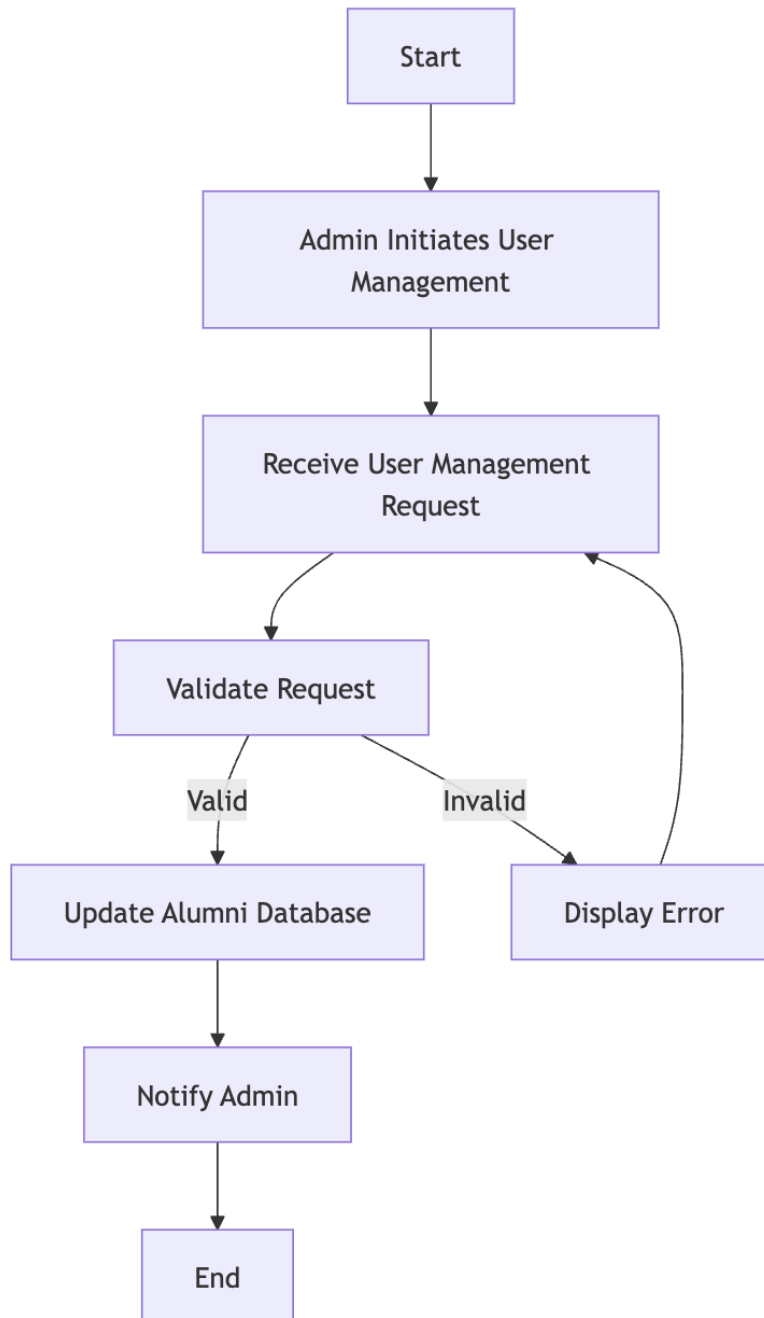


Figure 6: Activity Diagram for Manage Users Process

3.5 Activity - Post/View Events Process

This diagram covers posting (input details, validate, store, publish) or viewing (retrieve, display) events, with error handling for invalid inputs, ending after completion.

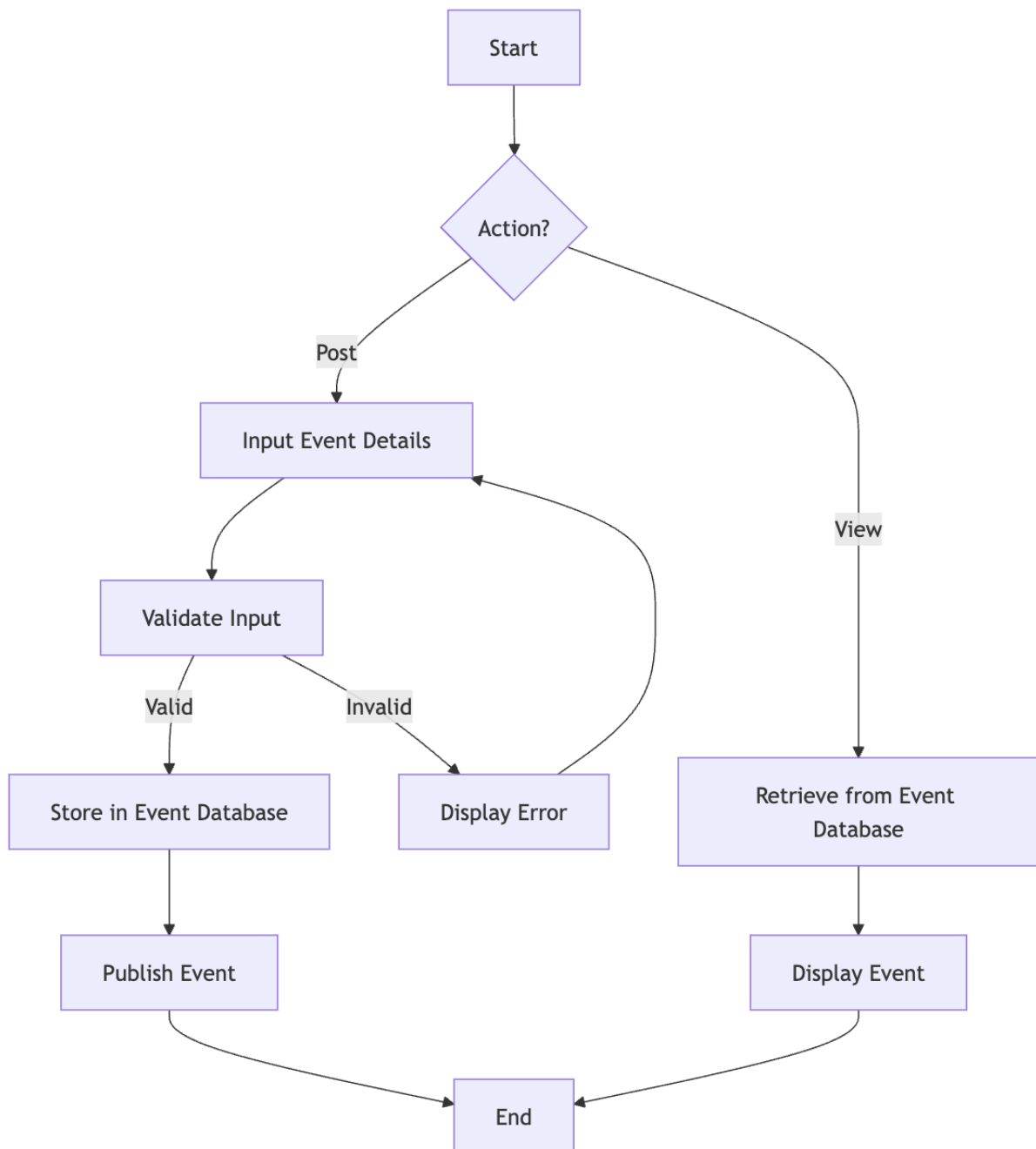


Figure 7: Activity Diagram for Post/View Events Process

4. Sequence - Subprocess Interactions

The following Sequence Diagrams illustrate the time-ordered interactions for each subprocess.

4.1 Sequence - Register Alumni Process

This diagram shows the interaction where an Alumni sends a registration request, the system validates it, stores valid data in the Alumni Database, notifies the Admin, and confirms with the Alumni. Invalid inputs return an error.

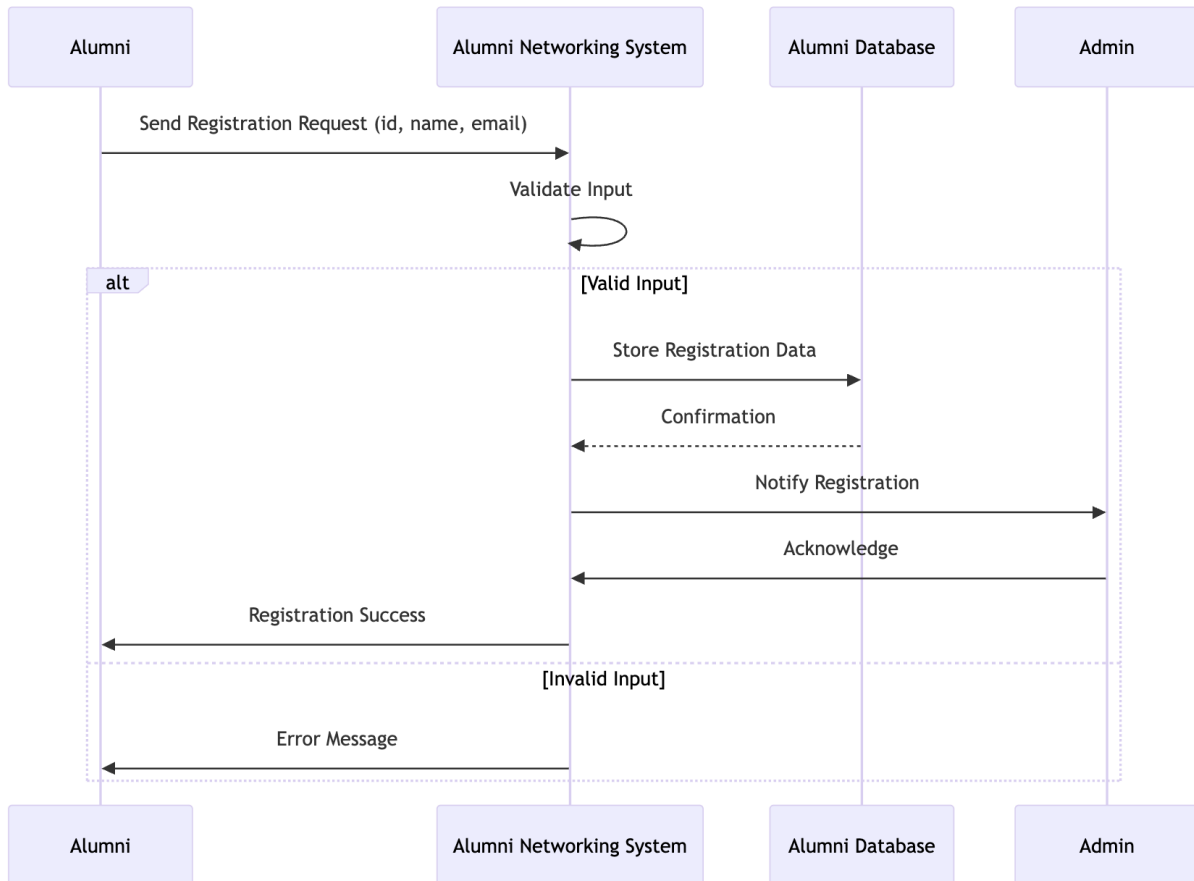


Figure 8: Sequence Diagram for Register Alumni Process

4.2 Sequence - Send Message Process

This diagram depicts a user sending a message, the system validating it, storing it in the Message Database, delivering it to the recipient, and confirming. Invalid messages return an error.

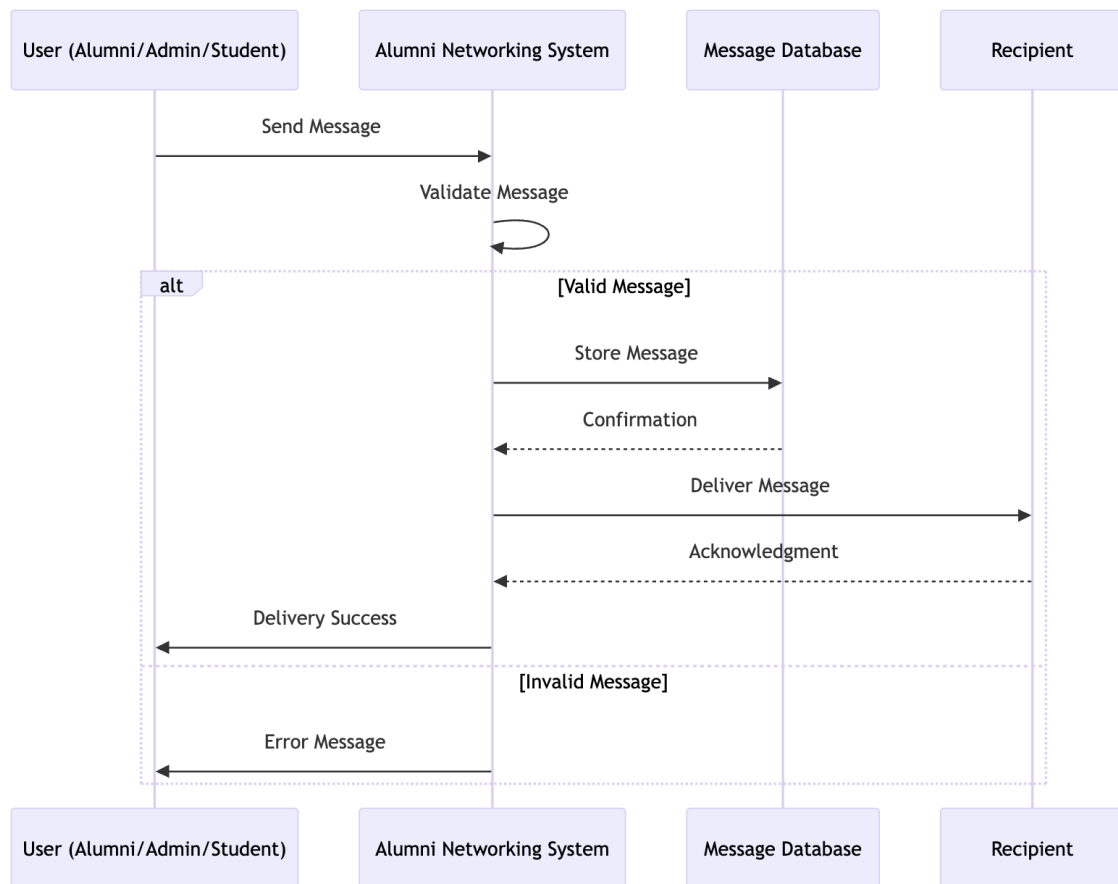


Figure 9: Sequence Diagram for Send Message Process

4.3 Sequence - Update Profile Process

This diagram illustrates an Alumni sending a profile update, the system validating it, updating the Alumni Database if valid, and confirming. Invalid updates return an error.

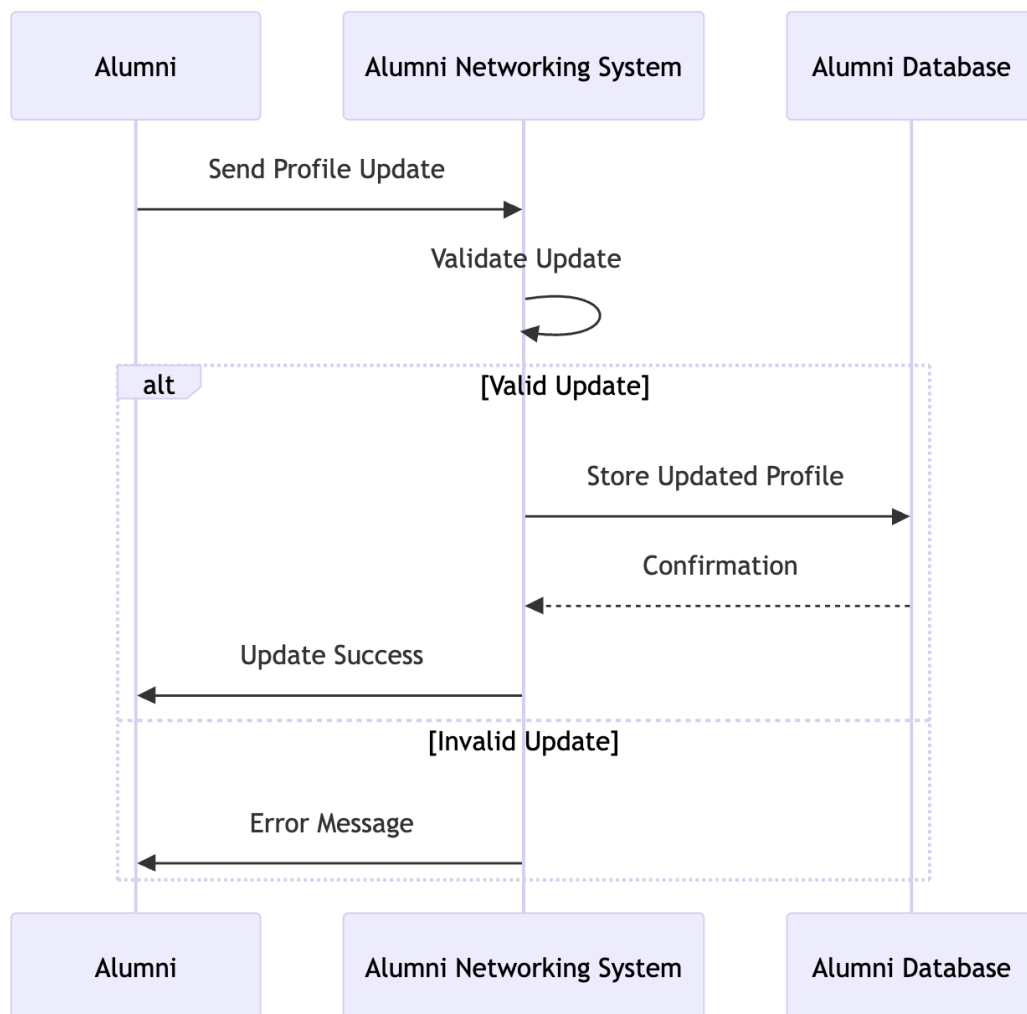


Figure 10: Sequence Diagram for Update Profile Process

4.4 Sequence - Manage Users Process

This diagram shows the Admin sending a user management request, the system validating it, updating the Alumni Database if valid, and confirming. Invalid requests return an error.

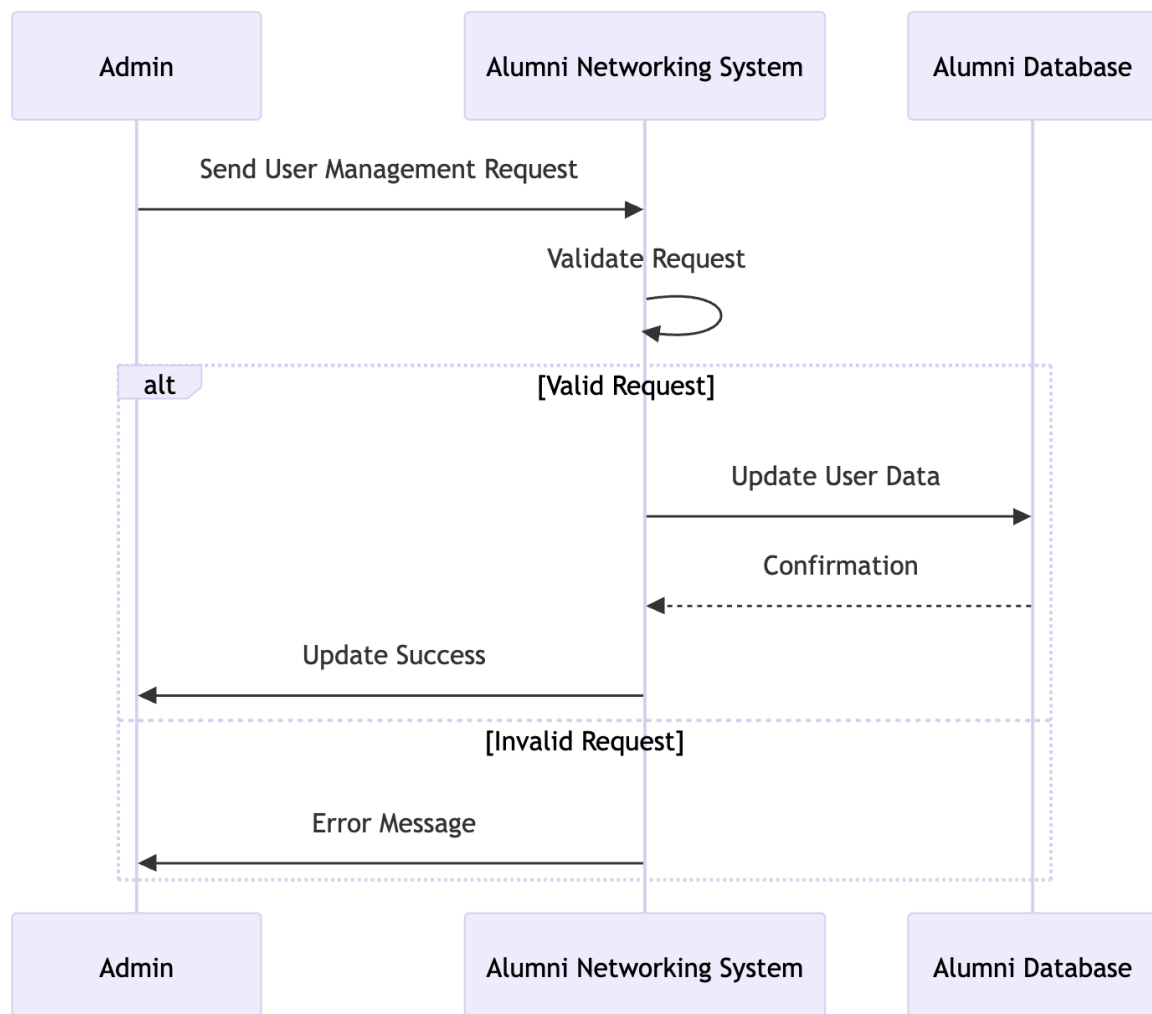


Figure 11: Sequence Diagram for Manage Users Process

4.5 Sequence - Post/View Events Process

This diagram handles posting (validate, store, publish) or viewing (retrieve, display) events, with conditional paths for valid/invalid inputs.

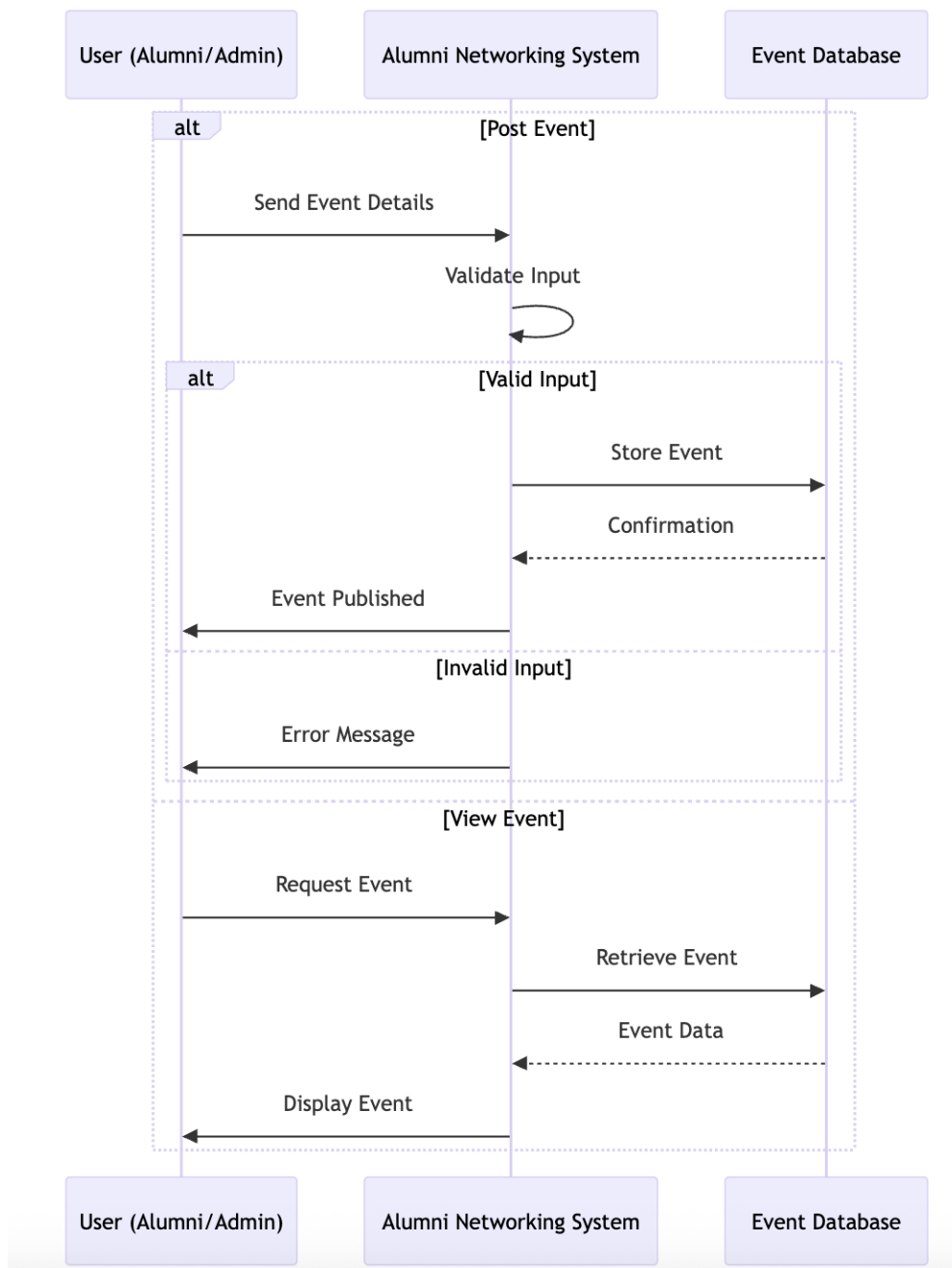


Figure 12: Sequence Diagram for Post/View Events Process

A Lab Report on Test Plan and Test Case for Web-Based Alumni Networking System



Submitted By:

Prachit Bhujel [PUR079BCT050]

Pratyos Panta [PUR079BCT054]

Sagar Mainali [PUR079BCT069]

Tribhuvan University

Institute of Engineering

Department of Electronics and Computer Engineering

Purwanchal Campus

Dharan-8, Sunsari

Objective

The primary aim of this lab session is to prepare a test plan and test case for the developed Web-Based Alumni Networking System to ensure all core functions work correctly and provide a reliable user experience.

Functional Overview

- **Test Plan:** Outlines the strategy, scope, and approach for testing.
- **Test Objectives:** Verify functionality, data integrity, bugs, and security/performance.
- **Test Cases:** Scenarios to validate system features.
- **Deliverables:** Test plan, cases, bug reports, and summary.

This report provides a structured testing framework to ensure system reliability.

Test Plan and Test Case

Test Plan

1. Introduction

This test plan outlines the strategy, scope, and approach for testing the Web-Based Alumni Networking System's core functions (registration, profile management, messaging, events, admin actions) to ensure reliability.

2. Test Objectives

- Verify login, profile, messaging, event management, and admin actions.
- Ensure database data integrity.
- Identify and fix UI/backend bugs.
- Validate security and performance.

3. Test Scope

In Scope:

- User registration/login.
- Profile creation/editing.
- Messaging between users.
- Event/announcement management.
- Admin user/content control.

Out of Scope:

- Third-party service integration.
- Load testing beyond 100 users.

4. Test Strategy

- Manual and automated testing (Selenium, Postman).
- Unit, integration, and user acceptance testing.
- Load and security testing for vulnerabilities.

5. Test Environment

- **OS:** Windows/Linux
- **Browser:** Chrome, Firefox, Edge
- **Server:** Localhost/Vercel
- **Database:** MongoDB
- **Backend:** Node.js/Express.js

6. Entry & Exit Criteria**Entry Criteria:**

- Application is deployed.
- Core modules are implemented.

Exit Criteria:

- Critical/high bugs fixed.
- All test cases passed/acceptable.

7. Test Deliverables

- Test Plan Document
- Test Case Document
- Bug Report
- Final Test Summary Report

Test Case

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
TC001	Valid User Registration	1) Enter valid email/password. 2) Submit.	User registered, redirected to login.	[To be filled]	Pass/Fail
TC002	Profile Creation	1) Go to profile. 2) Add name/bio. 3) Save.	Profile created, visible.	[To be filled]	Pass/Fail
TC003	Messaging Functionality	1) User A sends msg to B. 2) Check B's inbox.	Message delivered to B.	[To be filled]	Pass/Fail
TC004	Event Creation by Admin	1) Go to admin panel. 2) Create event. 3) Publish.	Event visible on dashboard.	[To be filled]	Pass/Fail

A Lab Report on Gantt and PERT Charts for Web-Based Alumni Networking System



Submitted By:

Prachit Bhujel [PUR079BCT050]

Pratyooos Panta [PUR079BCT054]

Sagar Mainali [PUR079BCT069]

Tribhuvan University

Institute of Engineering

Department of Electronics and Computer Engineering

Purwanchal Campus

Dharan-8, Sunsari

Objective

The primary aim of this lab session is to prepare Gantt and PERT charts for the Web-Based Alumni Networking System to visualize the testing schedule and task dependencies.

Functional Overview

- **Gantt Chart:** Provides a timeline visualization of testing tasks.
- **PERT Chart:** Maps dependencies and critical paths for testing.
- **Purpose:** Ensures efficient planning and risk mitigation.

This report uses Gantt and PERT charts to enhance the management of the testing phase.

Gantt and PERT Charts

1. Gantt Chart - Testing Timeline Visualization

The Gantt chart offers a clear, time-based visual representation of the testing schedule for the Web-Based Alumni Networking System. It breaks the testing phase into distinct tasks such as unit testing, integration testing, user acceptance testing, and documentation. Each task is depicted with: - Start and end dates - Task duration - Overlapping or dependent activities

In this context, the Gantt chart helps track testing progress, ensures timely task completion, and facilitates team coordination by identifying critical and parallel testing activities.

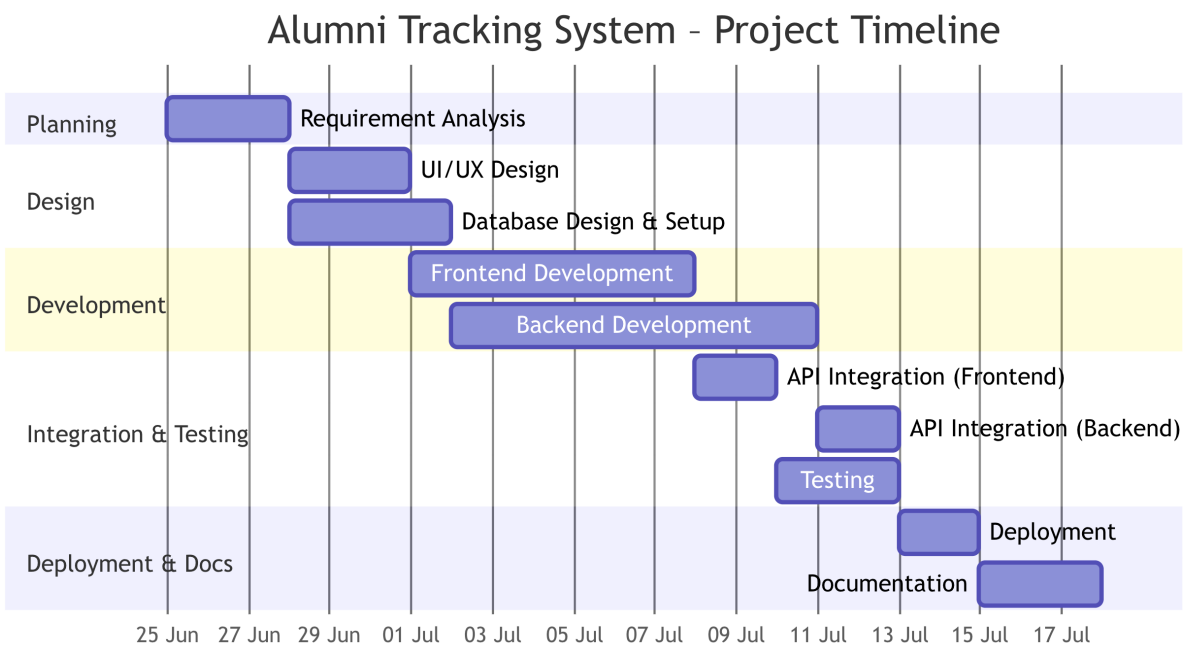


Figure 1: Gantt Chart for Testing Schedule

2. PERT Chart - Testing Dependency Mapping

The PERT (Program Evaluation and Review Technique) chart illustrates the dependencies and workflow of the testing phase for the Web-Based Alumni Networking System. It shows how each task is logically connected, with nodes representing tasks and directed edges indicating the sequence of execution. Key aspects include: - Sequential flow from unit testing to documentation - Parallel testing paths like integration and security testing - Critical path identification to minimize project delays

This chart is crucial for identifying task dependencies, estimating testing duration, and anticipating bottlenecks, thus aiding in risk mitigation and efficient resource allocation.

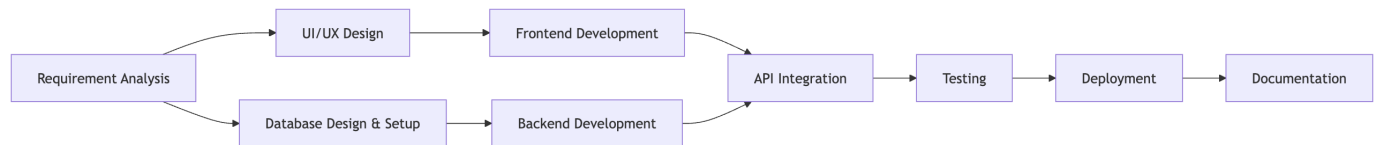


Figure 2: PERT Chart for Testing Dependencies